Job Aggregation Platform

Final Project Report

Prepared for: Engineering & Product Management

Prepared by: Prerna Anand

Date: July 6, 2026

Dataset analyzed: jobs.json — 155,794 unique job records

Source documents: research/api_analysis.md, reports/analytics_summary.md

Table of Contents

2. Executive Summary

The Job Aggregation Platform aggregates 155,794 unique job postings from 20 registered data providers into a single normalized schema, deduplicated by URL, and served through a static frontend. Nineteen of the 20 providers are currently contributing data; one (TheirStack) is blocked by an account plan restriction unrelated to the codebase.

Coverage is broad but concentrated. The United States accounts for 51% of all jobs, and a single provider — SmartRecruiters — supplies 41.8% of the entire dataset. A single employer, Domino's, alone accounts for 15.7% of every job in the platform, because its board lists each store-level opening as a separate posting across thousands of locations.

Deduplication is functioning correctly: zero duplicate URLs remain in the dataset. A large volume of same-title, same-company postings initially appears duplicate-like, but inspection confirms these are legitimate multi-location roles (retail, logistics, delivery) rather than aggregation errors — a naive dedup rule would have destroyed tens of thousands of real job listings.

Freshness is moderate but degrading. At collection time, 54.0% of jobs were posted within the prior 30 days; 46.0% are already stale, and 14.0% are over a year old. The pipeline has no scheduled refresh and no expiry mechanism, so this ratio will continue to worsen without intervention.

The highest-value opportunities identified are: (1) extending the existing, already-keyed Adzuna integration to additional countries — unlocking Canada, the UK, continental Europe, and Latin America with no new integration work; (2) introducing a scheduled refresh and expiry cycle to fix freshness; (3) closing the Singapore/Southeast Asia and India coverage gaps via a no-auth government API and existing ATS infrastructure; and (4) turning on date and location filters already supported by several integrated APIs but currently unused. None of these require scraping, new commercial contracts, or code changes to the aggregation pipeline's core logic.

3. Project Overview

The platform is a modular Python pipeline (jobs.py plus a sources/ package) that fetches job postings from every registered provider, normalizes each record into a common seven-field schema, deduplicates the combined list by URL, and writes the result to jobs.json. A lightweight static frontend (HTML/CSS/JavaScript) reads that file directly to render a searchable, filterable job board.

Schema

Every job record — regardless of source — is normalized to exactly seven fields: title, company, location, url, tags, remote, and posted (an ISO YYYY-MM-DD date string). This uniform shape is what allows 20 structurally different provider APIs to feed a single downstream pipeline and frontend.

Provider architecture

Providers were added in three phases: Phase 1 (6 no-authentication public APIs), Phase 2 (8 API-key-based providers, credentials loaded from a git-ignored .env file), and Phase 3 (6 company-based ATS integrations — Greenhouse, Lever, Ashby, SmartRecruiters, Workable, and Teamtailor — each fetching one job board per individually verified company). Every provider implements the same two-method interface (fetch_raw, normalize), so adding a new source never requires touching the core pipeline.

Reliability

Each provider's run() method isolates failures: a network error, bad response, or per-record normalization issue in one source is logged and skipped rather than aborting the entire run. Retryable HTTP statuses (429, 500, 502, 503, 504) are retried with exponential backoff before a request is given up on.

4. Dataset Overview

The current dataset (jobs.json) contains 155,794 unique job records spanning 9,497 distinct companies. Schema completeness is very high: only 45 records (0.03%) are missing a company name, only 72 (0.05%) are missing a location, and 155,779 records (99.99%) carry a parseable posting date.

Metric	Value
Total unique jobs	155,794
Registered providers	20
Providers contributing in latest run	19
Distinct companies	9,497
Jobs with empty company	45 (0.03%)
Jobs with empty location	72 (0.05%)
Jobs with empty tags	11,762 (7.5%)
Jobs with a parseable posted date	155,779 (99.99%)
Duplicate URLs remaining	0

Remote vs. onsite

Only 10.9% of jobs (16,992) are marked remote; the remaining 89.1% (138,802) are onsite or not marked remote. Several providers are remote-only by construction (Himalayas, RemoteOK, Jobicy); most others derive the remote flag from a genuine API field where available, falling back to a keyword heuristic on title/location otherwise.

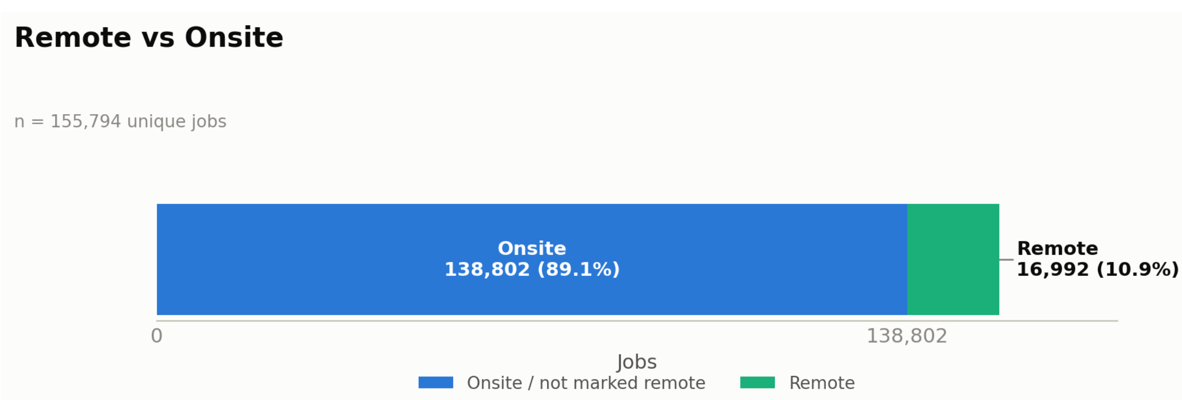

Figure 1. Remote vs. onsite split across the full dataset.

Company distribution

Company concentration is extreme and bimodal. The top 10 companies account for 37.8% of the dataset (58,922 jobs); the top 25 account for 47.5%. At the same time, 58.0% of all 9,497 distinct companies (5,505 companies) contribute exactly one job each. Domino's alone represents 15.7% of every job in the platform — nearly one in six listings — because its SmartRecruiters board lists each store-level opening (delivery driver, assistant manager, etc.) as a separate posting across thousands of U.S. locations.

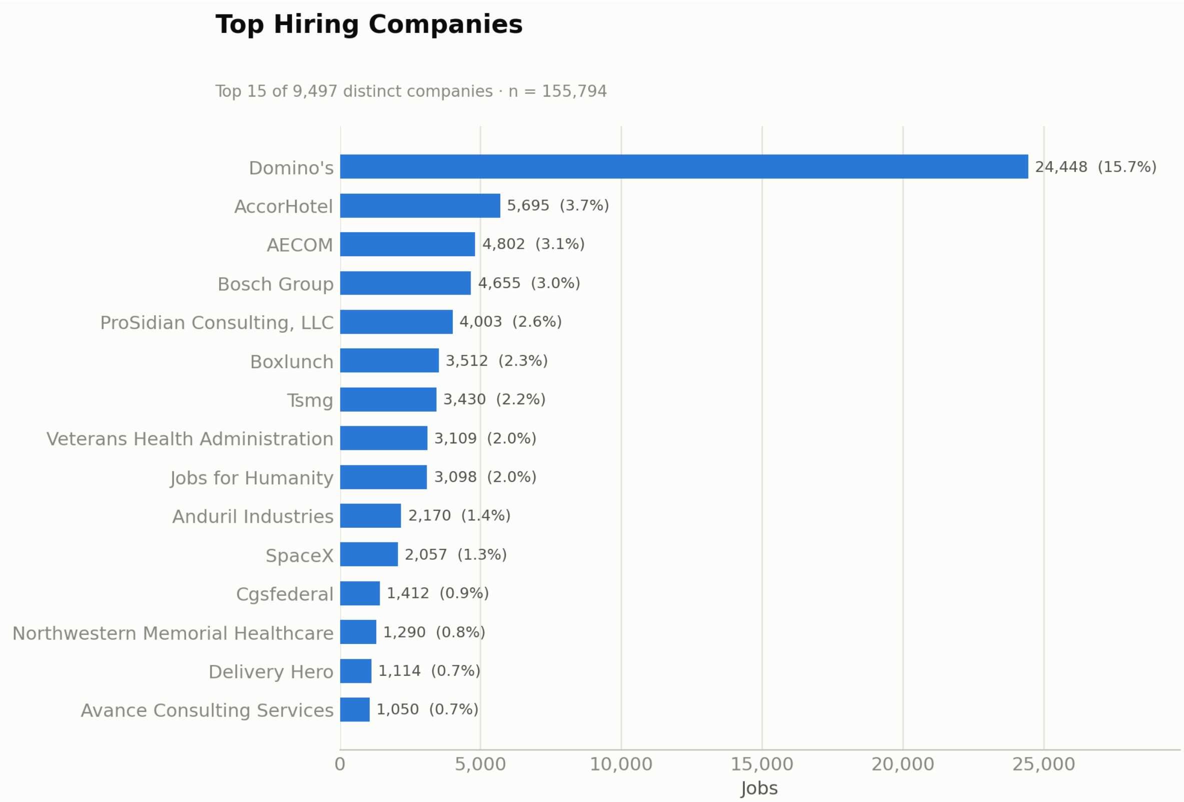

Figure 2. Top 15 hiring companies by job count.

5. Provider Analysis

Provider attribution was reconstructed from each job's URL host, since the schema does not carry an explicit source field. A single provider, SmartRecruiters, supplies 41.8% of the entire dataset — more than four in ten jobs. The top three providers (SmartRecruiters, Lever, Greenhouse) together account for 62.8% of all jobs.

Provider	Jobs	% of dataset
SmartRecruiters	65,193	41.8%
Lever	17,755	11.4%
Greenhouse	14,930	9.6%
USAJOBS	10,000	6.4%
Reed	9,000	5.8%
Bundesagentur (+AMS)	~7,800	~5.0%
Ashby	5,171	3.3%
Workable	4,159	2.7%
Jooble	~3,100	~2.0%
Himalayas	2,964	1.9%
Adzuna	2,001	1.3%
The Muse	2,000	1.3%
Arbeitnow	930	0.6%
Teamtailor	474	0.3%
Other (SerpApi, OpenWeb Ninja, Careerjet, RemoteOK, Jobicy)	~350	~0.2%
TheirStack	0	0.0% (plan-gated, HTTP 402)

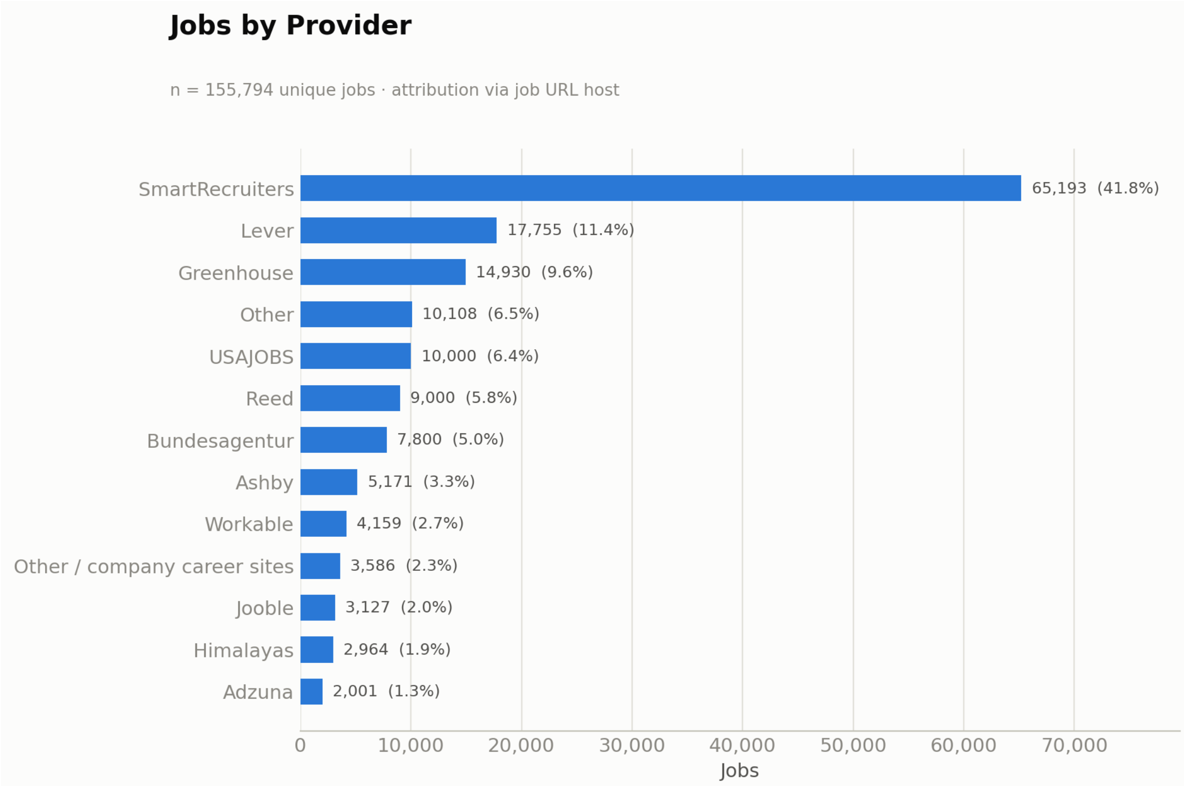

Figure 3. Jobs by provider (top 12, remainder grouped as Other).

ATS provider mix

The six company-based ATS providers (Greenhouse, Lever, Ashby, SmartRecruiters, Workable, Teamtailor) together supply 107,664 jobs — 69.1% of the entire dataset. Within that ATS-sourced subset, SmartRecruiters alone represents 60.6%, followed by Lever (16.5%) and Greenhouse (13.9%); Ashby, Workable, and Teamtailor together make up the remaining 9.1%.

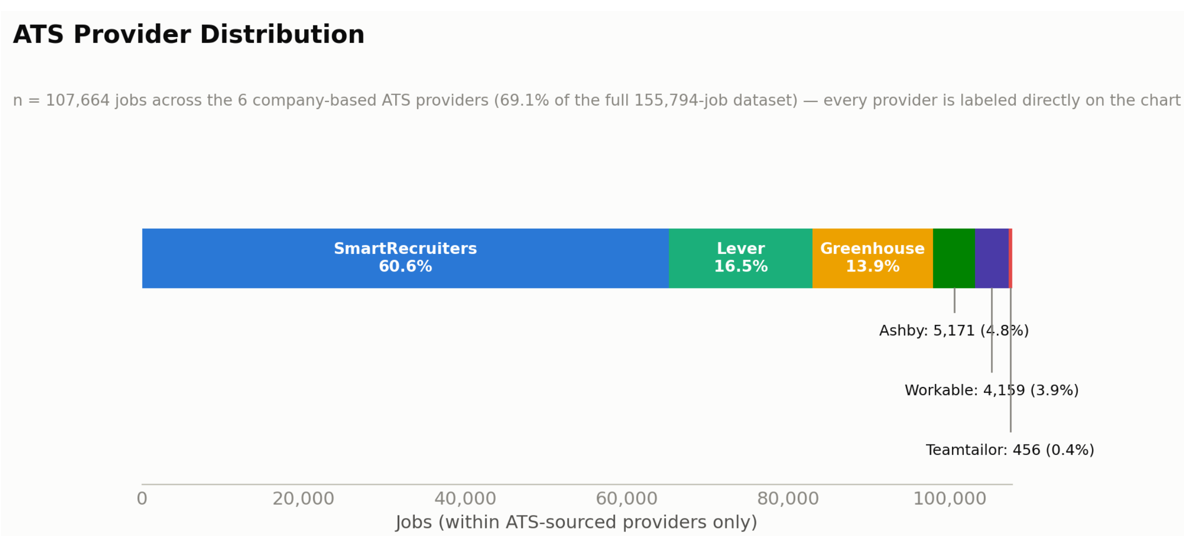

Figure 4. Distribution across the six ATS providers only.

All six ATS providers grow exclusively by manually adding verified company slugs to a curated list — there is no query-based way to expand their reach. Regional and company composition is therefore driven largely by which companies were curated into these lists, not by a balanced global crawl.

6. Regional Coverage

Region was inferred from each job's free-text location field using a keyword/token classifier. The United States dominates at 51.0% of the dataset; the six primary target regions together account for 81.3% (126,638 jobs).

Region	Jobs	% of dataset
United States	79,427	51.0%
Europe (continental, non-UK)	21,886	14.0%
United Kingdom	10,305	6.6%
LATAM	6,955	4.5%
Singapore / Southeast Asia	4,629	3.0%
Canada	3,436	2.2%
Remote / Worldwide	822	0.5%
Other / Unclassified	28,334	18.2%

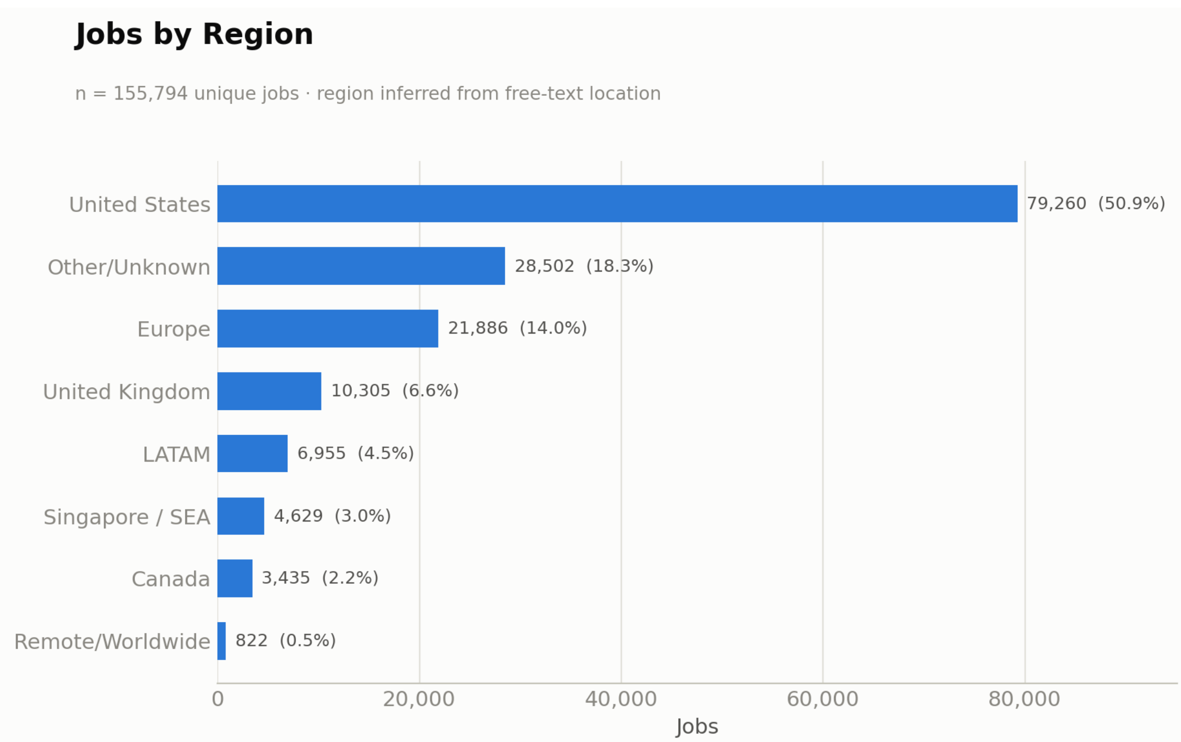

Figure 5. Jobs by region.

Canada is the weakest of the six primary target regions at 2.2% — under half the size of the UK's dedicated coverage. 18.2% of jobs carry location text too ambiguous to classify into a region, which limits the reliability of any location-based filter, count, or map feature in its current form.

Top countries

Rank	Country	Jobs	%
1	United States	79,427	51.0%
2	Germany	12,167	7.8%
3	United Kingdom	10,305	6.6%
4	India	3,471	2.2%
5	Canada	3,436	2.2%
6	France	2,218	1.4%
7	China	1,435	0.9%
8	Australia	1,225	0.8%

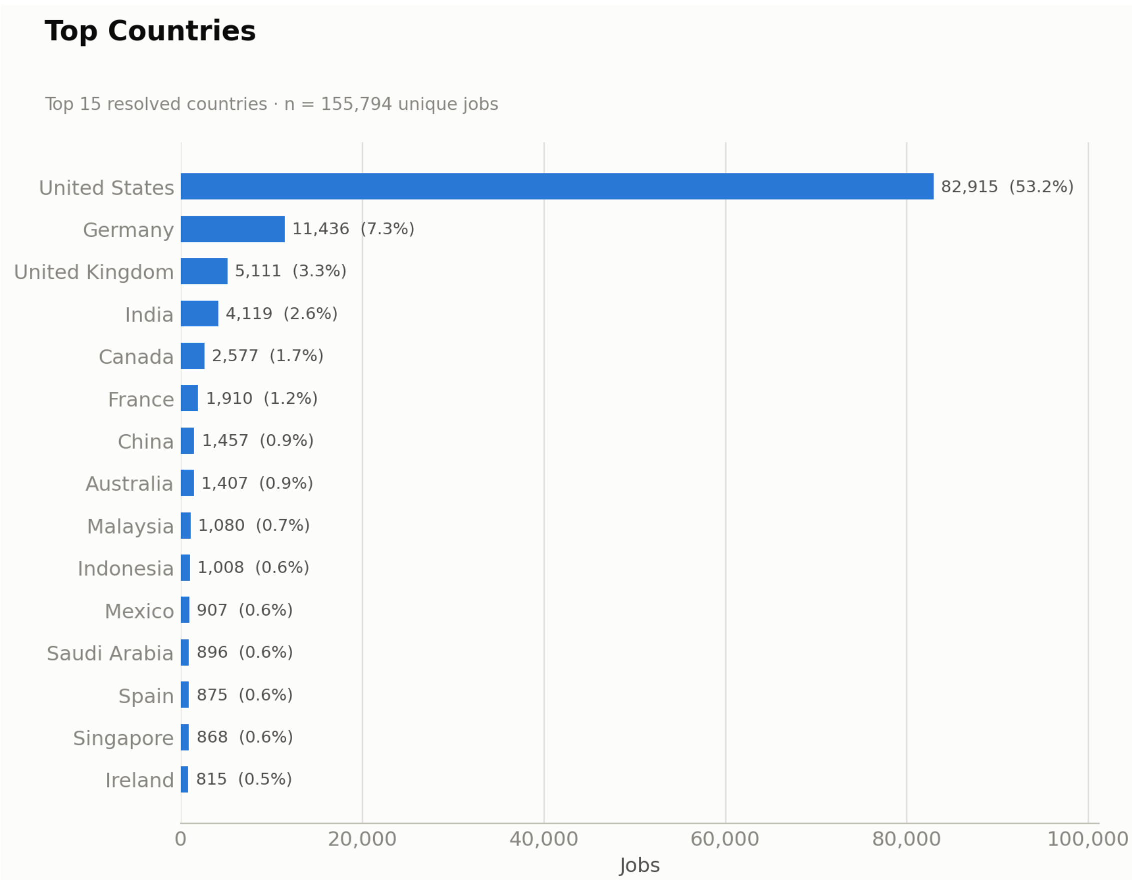

Figure 6. Top 15 resolved countries by job count.

Europe is largely a Germany story: Germany's 12,167 jobs represent 56% of the entire European total, driven by the two German public-agency sources. India, at 2.2% (3,471 jobs), is already the fourth-largest country in the dataset despite having no dedicated source — see Section 10.

7. Job Freshness

Age is measured against the dataset's collection date (2026-07-03, the newest posted value present in the data). At collection time, a slim majority of the dataset was current, but the stale fraction is already substantial and will only grow without a refresh mechanism.

Age at collection	Jobs	Cumulative 'within'	Cumulative %
Within 24 hours	24,052	24,052	15.4%
Within 3 days	+6,582	30,634	19.7%
Within 7 days	+11,884	42,518	27.3%
Within 14 days	+24,610	67,128	43.1%
Within 30 days	+16,949	84,077	54.0%
Older than 30 days	71,702	—	46.0%
Unknown / unparseable	15	—	0.0%

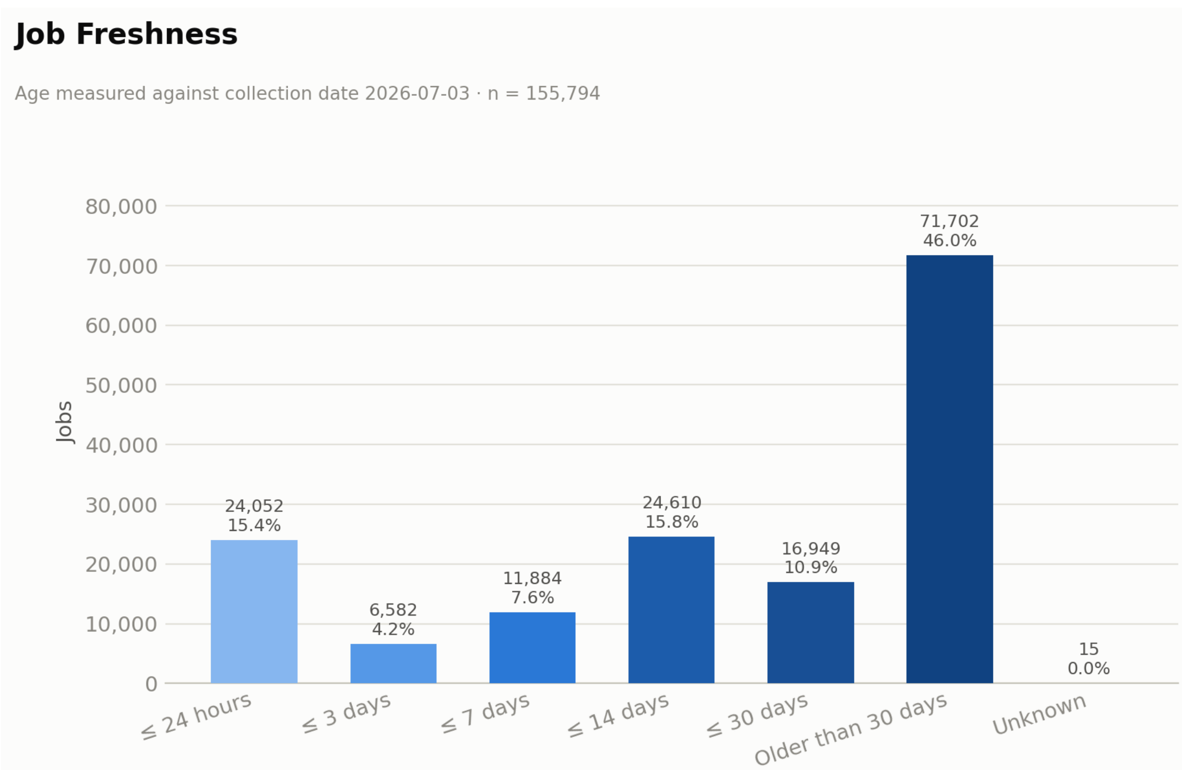

Figure 7. Job freshness distribution (buckets from freshest to stalest).

Total current jobs (≤ 30 days): 84,077 — 54.0% of the dataset. Total stale jobs (> 30 days, including unknown): 71,717 — 46.0%.

Deep-stale tail

Age	Jobs	%
Older than 90 days	45,075	28.9%
Older than 180 days	29,640	19.0%
Older than 365 days	21,768	14.0%

The oldest posting on record dates to 2009-12-05; the median posting date is 2026-06-09. The stale tail is concentrated in large enterprise ATS boards, which return every currently open requisition, including long-standing evergreen roles. Because the dataset is a static snapshot with no scheduled refresh and no expiry logic, the current/stale ratio degrades every day after collection with no offsetting mechanism.

8. Duplicate Analysis

The pipeline deduplicates by exact URL, and this is functioning correctly: zero duplicate URLs and zero empty URLs remain in the dataset. Under looser candidate keys, however, a large number of same-title/company rows appear to “collide” — this section quantifies that finding and explains why it is not, in fact, a duplication problem.

Check	Result
Exact-URL duplicates remaining	0
Jobs with empty URL	0
Title + company collisions (extra rows)	35,300 across 11,340 groups
Title + company + location collisions (extra rows)	8,290 across 4,963 groups
Title+company groups spanning more than one distinct URL	11,340 (100% of those groups)

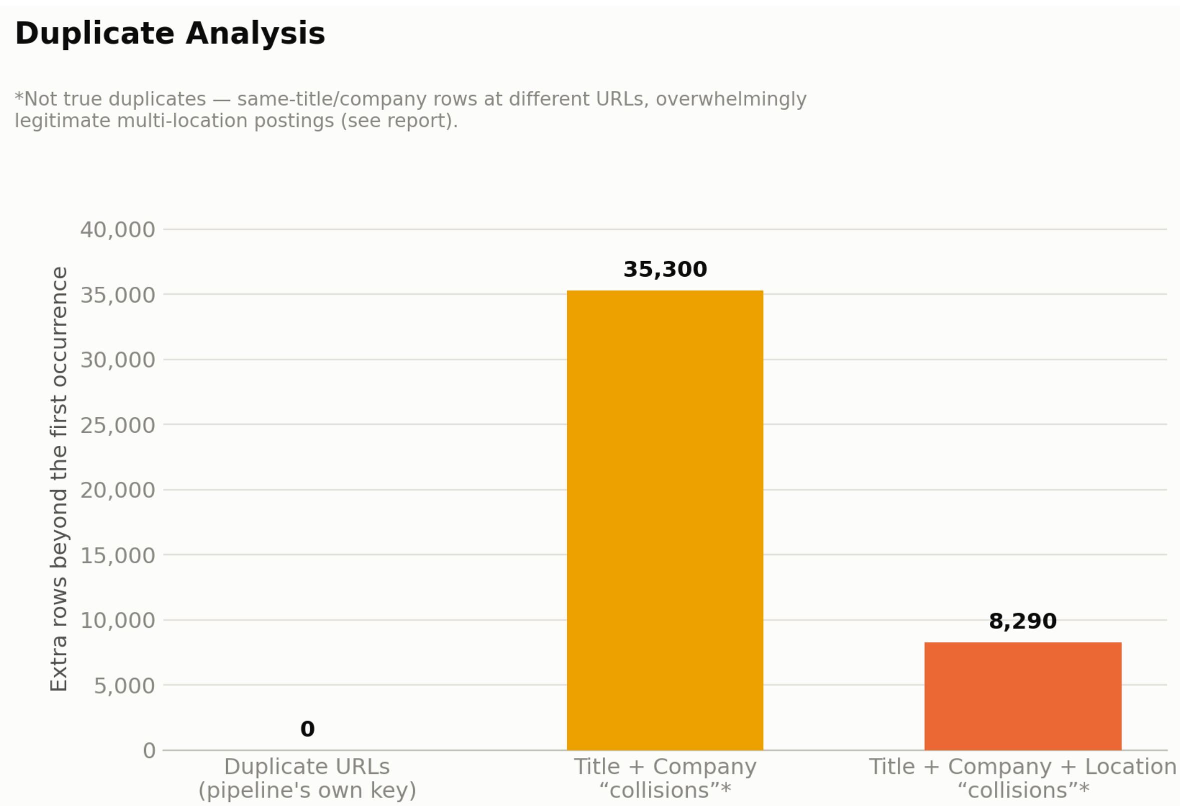

Figure 8. Duplicate counts under three candidate keys.

Critically, every one of the 11,340 title+company groups spans multiple distinct URLs — these are not missed duplicates. Inspection shows they are overwhelmingly the same role posted at hundreds of separate physical locations: Domino's “Delivery Driver,” BoxLunch “Sales Associate,” and similar multi-location retail/logistics roles. Applying a naive title+company deduplication rule would have erased tens of thousands of legitimate, distinct job openings.

The real, and much smaller, duplicate risk is cross-provider overlap between aggregator-style sources (e.g., Jooble, Careerjet, and Adzuna occasionally re-listing the same third-party posting) — a risk the current URL-keyed dedup does not directly measure and any future duplicate-reduction effort should target conservatively, without collapsing on title+company alone.

9. API Analysis

All 20 registered providers were audited for authentication model, rate and result-window limits, pagination behavior, and available filters. Three sources are already capped by the provider's own hard result window (USAJOBS and Bundesagentur at 10,000 jobs; Reed near its ~9,900-job ceiling). Three more are deliberately throttled to control cost on metered/paid APIs (SerpApi, OpenWeb Ninja, and TheirStack, each self-capped around 50 jobs per run).

Priority	Providers	Rationale
Critical	SmartRecruiters, Adzuna	SmartRecruiters is already the volume backbone; Adzuna is the single highest-leverage untapped asset (one keyed integration can add several regions by iterating its country parameter).
High	Lever, Greenhouse, USAJOBS, Reed, Bundesagentur	Large, free/near-free backbones for major regions, though each is bounded by a curated company list, federal-only scope, or a hard result window.
Medium	Ashby, Workable, Himalayas, The Muse, Jooble, Careerjet	Solid contributors with real breadth but individually smaller or with untapped upside (e.g., Himalayas' richer search endpoint).
Low	Arbeitnow, RemoteOK, Jobicy, Teamtailor, SerpApi, OpenWeb Ninja, TheirStack	Small absolute volume or deliberately throttled metered sources; TheirStack currently contributes zero jobs.

Full per-provider detail — authentication, pricing, rate limits, pagination, and every supported filter — is documented in research/api_analysis.md, Sections 4 and 9.

10. India Research

India is already the 4th-largest country in the dataset (3,471 jobs, 2.2%) despite having no dedicated source. A comprehensive survey of the Indian recruitment ecosystem — government portals, private job boards, ATS/HR platforms, staffing firms, and regional/niche boards — found a consistent structural pattern: India has a large, active online hiring market but almost no aggregatable public job APIs.

Key findings

- Government: the National Career Service (NCS) publishes vacancy data via data.gov.in as a periodic open dataset, not a live searchable API — authoritative but freshness-limited.
- Private boards: Naukri, Foundit, Apna, Shine, TimesJobs, Instahyre, Hirist, iimjobs, Internshala, Unstop, Wellfound, and LinkedIn/Indeed India all expose no official public read API.
- India-origin ATS/HRMS platforms (Zoho Recruit, Freshteam, Darwinbox, Keka, GreytHR) are uniformly per-tenant — useful only to that specific employer's own integration, not aggregatable across employers.
- Major Indian staffing firms (TeamLease, Quess, Randstad India, etc.) expose no public job feeds.

Recommended routes (ranked)

Rank	Route	Effort	ROI
1	ATS slug expansion for Indian-HQ companies on Greenhouse/Lever/Ashby/Workable/SmartRecruiters (already integrated)	Low	High
2	India-scoped query on existing metered sources (JSearch/OpenWeb Ninja, SerpApi)	Low	High
3	Paid aggregator (Fantastic Jobs / Active Jobs DB) covering Indian ATS and boards	Low	Medium (budget-gated)
4	National Career Service open dataset (data.gov.in)	Medium (ETL)	Medium
5	Enterprise data providers (Coresignal, JobsPikr)	Medium	Low (cost)
—	Scraping Naukri / Foundit / Apna / LinkedIn / Indeed	—	Not recommended (legal/maintenance risk)

Bottom line: legal, low-maintenance India coverage comes from extending infrastructure already built for this project (ATS slugs and existing metered/Google-Jobs sources), not from integrating a new Indian ATS vendor or scraping consumer boards.

11. Additional Regional APIs

A verified market survey was conducted across Canada, the UK, Europe, Singapore, Southeast Asia, and LATAM to identify additional free or public data sources beyond the 20 already integrated.

Region	Best candidate	Effort	Priority
Canada	Adzuna ‘ca’ (existing key, config-level change)	Trivial	Critical
United Kingdom	Adzuna ‘gb’ (de-risks single-source dependency on Reed)	Trivial	High
Europe	Adzuna multi-country (fr, de, es, nl, it, at, pl)	Trivial	Critical
Europe	France Travail (official, free, OAuth2 registration)	Medium	High
Singapore	MyCareersFuture — official, no-auth public API (~80k jobs)	Low	Critical
Southeast Asia	ATS slug expansion for SEA-headquartered companies	Low	Medium
LATAM	Adzuna ‘br’ + ‘mx’ (existing key)	Trivial	Critical
LATAM	Get on Board — the one LATAM board with a real public API	Low–Medium	Investigate

A separate verification pass explicitly ruled out 20 well-known regional boards — Eluta, Workopolis, Jobillico, Talent.com, WowJobs (Canada); TotalJobs, CV-Library, Jobsite, CWJobs (UK); StepStone, Xing Jobs, Jobrapido, Welcome to the Jungle (Europe); Computrabajo, Bumeran, OCC Mundial (LATAM); and JobStreet, JobsDB, Kalibrr, Glints (Southeast Asia) — all of which are either closed with no public API, or offer only a paid partner/posting channel rather than a free read feed. Get on Board (LATAM) was the sole exception found with a genuine public API.

The headline finding: the Adzuna key the project already holds is currently querying only the United States. Extending it to additional countries is, by a wide margin, the single highest-leverage change available — it touches five of the six target regions without any new integration work.

12. API Filter Optimization

An audit of all 20 integrations found that most run deliberately unfiltered “browse everything” queries — the right choice for a first pass, but one that leaves three significant levers unused.

- Date filters (e.g., Adzuna max_days_old, USAJOBS DatePosted, Jooble datecreatedfrom) are supported by at least six integrated providers but used by none — turning them on would cut the stale tail at ingest rather than after the fact.
- Country/locale filters (Adzuna's country path, Careerjet's locale_code, Himalayas' country parameter) are supported but mostly unused — turning them on directly targets the weak regions identified in Section 6.
- Query partitioning — splitting one broad query into several filtered ones by category, location, or date range — is the only way to exceed the hard result-window caps already reached by USAJOBS (10,000), Bundesagentur (10,000), Reed (~9,900), The Muse (2,000), Jooble (~1,100), and Himalayas (3,000 of a 90,000-job archive). This is the single largest coverage lever available without adding any new provider.

Rough planning estimate: turning on these unused filters and partitioning the capped sources could add approximately 40,000 to 90,000 additional useful, fresher, and less-duplicated jobs without integrating a single new provider. Full per-provider filter detail is in research/api_analysis.md, Section 17.

13. Maximum Coverage from Existing Providers

Every one of the 20 registered providers was audited individually to answer a narrower question than the rest of this report: is each provider we already have wired up already returning everything it realistically can, given its own API's own rules — and if not, how much more is obtainable using only capabilities that provider's API already exposes?

Are we already extracting the maximum realistic number of jobs from each provider?

For 17 of the 20 providers, no — meaningful additional volume is available using capabilities already present in the provider's own API, without any new authentication, scraping, or commercial agreement. Only three providers (Arbeitnow, RemoteOK, and TheirStack) are already at their practical ceiling, and one of those three (TheirStack) is capped by an account billing restriction rather than by the API itself.

Which providers still have untapped potential?

- Adzuna — by far the largest opportunity in the entire provider set. The current 2,000-job figure reflects querying a single country (the US) out of 10+ countries the existing API key already has access to.
- Bundesagentur, USAJOBS, and Reed — each already sits at or near a hard per-query result-window ceiling (10,000, 10,000, and ~9,900 respectively), but each also supports a location or keyword filter that can open an independent window of similar size when the query is partitioned instead of run once, broadly.
- Himalayas, The Muse, Jooble, and Careerjet — each is currently run as a single broad or wildcard query well below its real ceiling; switching to filtered, partitioned queries (by country, category, or real keywords instead of a wildcard) would meaningfully increase volume.

Which providers have already reached their practical limits?

- Arbeitnow — the entire current board is already fetched to exhaustion via full pagination, and the API offers no keyword, location, or category parameter to partition further.
- RemoteOK — the documented public endpoint is a fixed ~100-job snapshot with no confirmed filtering or pagination mechanism.
- TheirStack — effectively at zero due to an account plan restriction (HTTP 402), not because its API has been exhausted; this should be reassessed the moment Jobs API access is restored on the account.

Which providers can be improved using existing API filters?

Eight providers — Adzuna, Bundesagentur, USAJOBS, Reed, Himalayas, Jooble, Careerjet, and The Muse — already support date, country/locale, category, or location filters that the current integration does not use. Turning these on, or partitioning a single broad query into several filtered ones, is a pure configuration and query-logic change with no new credentials required.

Which providers require adding additional companies rather than changing code?

All six ATS providers — Greenhouse, Lever, Ashby, SmartRecruiters, Workable, and Teamtailor — expose no keyword, location, category, or date request filter at all; each is a full-board, per-company fetch. For these six, the API itself is already fully exploited for every currently configured company, so the entire opportunity lies in expanding the curated company-slug list, not in changing how the API is queried.

Which providers cannot realistically return more jobs due to API limitations?

Arbeitnow and RemoteOK. Both expose a complete, unfilterable snapshot of their respective boards with no query dimension to expand beyond what is already being collected — any further volume would depend on the boards themselves growing over time, not on anything this project can change.

Provider-by-provider comparison

Provider	Current Jobs Collected	Estimated Practical Maximum	Additional Jobs Possible	Recommended Optimization Strategy
Arbeitnow	930	~950–1,000	~0–100	None — full board already exhausted
Himalayas	3,000	13,000–23,000	+10,000–20,000	Switch to richer search endpoint; partition by country/category
RemoteOK	100	~100	~0	None confirmed; verify tag-scoped endpoints before assuming more
Jobicy	100	300–500	+200–400	Partition by geo/industry/tag across multiple calls
The Muse	2,000	5,000–8,000	+3,000–6,000	Partition by category/location; consider an API key
Bundesagentur	10,000	25,000–40,000	+15,000–30,000	Partition by region or keyword
Jooble	1,129	4,000–9,000	+3,000–8,000	Replace wildcard with targeted keywords/locations
USAJOBS	10,000	20,000–30,000	+10,000–20,000	Partition by job category or location
Adzuna	2,000	22,000–42,000	+20,000–40,000	Query additional countries already covered by the existing key
Reed	9,000	17,000–24,000	+8,000–15,000	Partition by UK region/keyword
SerpApi	50	~2,500/mo (budget-bound)	Budget-dependent	Reallocate quota to targeted location/date queries
OpenWeb Ninja	~47	~200/mo (budget-bound)	Budget-dependent	Reallocate quota toward country-scoped (e.g. India) queries
Careerjet	2,000	7,000–12,000	+5,000–10,000	Add locale-scoped passes + real keywords
TheirStack	0	0 (plan-gated)	0 until plan restored	Resolve the account/billing restriction first
Greenhouse	22,728	27,728–37,728	+5,000–15,000	Expand curated company-slug list
Lever	17,755	22,755–27,755	+5,000–10,000	Expand curated company-slug list
Ashby	5,171	7,171–10,171	+2,000–5,000	Source additional slugs outside AI/SaaS/FinTech
SmartRecruiters	65,193	70,193–80,193	+5,000–15,000	Expand slug list, but weigh against concentration risk
Workable	7,644	10,644–14,644	+3,000–7,000	Source slugs in staffing/consulting/fintech/healthcare
Teamtailor	474	974–1,974	+500–1,500	Source additional career sites (highest candidate hit rate)

Summary

- Highest ROI providers to optimize first: Adzuna (+20,000–40,000, no new integration, just additional countries on the existing key), followed by Bundesagentur, USAJOBS, Reed, Himalayas, Jooble, and Careerjet — together representing roughly 56,000–99,000 further jobs from filters and query partitioning these APIs already support.
- Providers already operating near their maximum: Arbeitnow and RemoteOK have no remaining query-based lever; TheirStack is at zero solely due to a billing restriction rather than an API ceiling.
- Expected increase in coverage before integrating additional APIs: on the order of 75,000–140,000 additional jobs are realistically obtainable from the eight already-integrated, already-credentialed providers with untapped filter capacity — before any new provider from the regional or India research in this report needs to be added.

14. Engineering Recommendations

- Broaden regional balance — particularly Canada, UK resilience, and Singapore/Southeast Asia — by extending already-integrated providers rather than adding new integrations.
- Introduce a scheduled refresh and expiry mechanism so the freshness ratio stops degrading between runs and closed postings are retired automatically.
- Grow ATS company-slug lists for under-represented regions (Canada, SEA, LATAM, India) to rebalance provider and regional concentration.
- Do not deduplicate on title+company alone — doing so would remove tens of thousands of legitimate multi-location jobs. Any future duplicate-reduction work should combine URL normalization with conservative, location-aware matching.
- Improve location-text normalization to shrink the 18.2% unclassifiable bucket and make regional filtering and reporting more reliable.
- Revisit the TheirStack account/plan to restore its contribution, or formally retire it from the active provider list if the restriction is permanent.
- Monitor concentration risk — with 41.8% of jobs from one provider and 15.7% from one company, any change to SmartRecruiters' Domino's board would materially shift dataset-wide statistics.
- Turn on already-supported date and country/locale filters (Section 12) before considering any new integration — this is free, low-risk, and high-impact.

15. Roadmap

Recommendations are grouped into five priorities, each with an estimated coverage increase, engineering effort, and business value.

Priority	Focus	Net-new jobs	Effort	Business value
P1	Adzuna multi-country + date filter	15,000–25,000	Low	Very High
P2	Freshness foundation (scheduling + expiry)	0 (quality fix)	Medium	Very High
P3	Weak regions: MyCareersFuture + ATS slugs + India query	40,000–70,000	Low–Medium	High
P4	Result-window bypass via partitioning + France Travail	20,000–50,000	Medium	High
P5	Dedup/location normalization + optional paid aggregator	Quality (+50,000 optional)	Medium	Medium–High

If only two weeks of engineering time were available

- Days 1–2: Ship Adzuna multi-country + max_days_old (Priority 1).
- Days 2–3: Capture a fetched_at timestamp and surface a “data as of” signal.
- Days 3–6: Build the scheduling + presence-diff expiry core (Priority 2).
- Days 6–9: Add MyCareersFuture (Singapore) and a first batch of Indian/SEA/Canada ATS slugs (Priority 3).
- Days 9–12: Prove out one result-window partition (e.g., Reed or The Muse) plus URL normalization in dedup.
- Days 12–14: Integration testing, dedup verification, and rate-limit/budget tuning across the newly added passes.

This sequence realistically adds ~40,000–70,000 net-new, better-distributed jobs and fixes freshness within two weeks, without any scraping or new commercial contracts.

16. Future Improvements

Beyond the two-week roadmap, several structural investments would compound the platform's value as the dataset grows:

- A persistent datastore (e.g., Postgres or OpenSearch) replacing the single jobs.json file, paired with a paginated, queryable search API — the current design of loading the entire file into the browser will not scale past the low hundreds of thousands of jobs.
- Incremental/delta ingestion with per-job first_seen/last_seen tracking and automated expiry, replacing the current full-overwrite snapshot model.
- Scalable deduplication — normalized-URL keys plus blocked fuzzy/locality-sensitive-hash matching — to handle cross-provider overlap at higher volume, and description-based matching once job descriptions are added to the schema.
- Automated ATS company/slug discovery and health monitoring, so coverage grows without linear manual curation effort.
- A decoupled, per-source pipeline with queueing and rate-limit budgeting, so one slow or large provider cannot stall an entire run, with observability on volume, freshness, and duplicate rate per source per run.
- Optional adoption of a paid aggregator (e.g., Fantastic Jobs' Active Jobs DB) for a step-change in both volume and freshness once budget allows.

17. Recent Job Market Analysis (Last 30 Days)

This section restricts analysis to the 84,077 jobs posted within 30 days of the dataset's 2026-07-03 collection date — 54.0% of the full 155,794-job dataset — the same <=30-day slice already established in Section 7 (Job Freshness). Jobs older than 30 days are excluded entirely from every figure below.

1. Job Category Distribution

Every recent job was classified into a business-friendly category from its title (and tags where informative). Because this dataset is heavily weighted toward retail, hospitality, food-service, and driving/logistics postings, a large share does not map cleanly onto a professional/technical taxonomy and is honestly reported as Other rather than forced into a poor-fitting bucket.

Category	Recent Jobs	% of recent
Other	44,858	53.4%
Supply Chain	7,531	9.0%
Software Engineering	5,571	6.6%
Customer Support	5,264	6.3%
Healthcare	3,505	4.2%
Sales	2,888	3.4%
Executive	1,570	1.9%
Education	1,474	1.8%
Accounting	1,224	1.5%
Project Management	994	1.2%
Internship / Graduate	911	1.1%
Manufacturing	792	0.9%
Human Resources	779	0.9%
Marketing	755	0.9%
Product Management	576	0.7%
Finance	525	0.6%
DevOps / Cloud	517	0.6%
Data Science / Analytics	511	0.6%
Legal	399	0.5%
Business Development	392	0.5%
Cybersecurity	374	0.4%
Administrative	334	0.4%
Customer Success	302	0.4%

Category	Recent Jobs	% of recent
AI / Machine Learning	299	0.4%
Digital Marketing	293	0.3%
Operations	274	0.3%
Procurement	272	0.3%
Recruiting	250	0.3%
IT Support	219	0.3%
UI / UX Design	174	0.2%
Government	102	0.1%
Pharmaceutical	77	0.1%
Graphic Design	71	0.1%

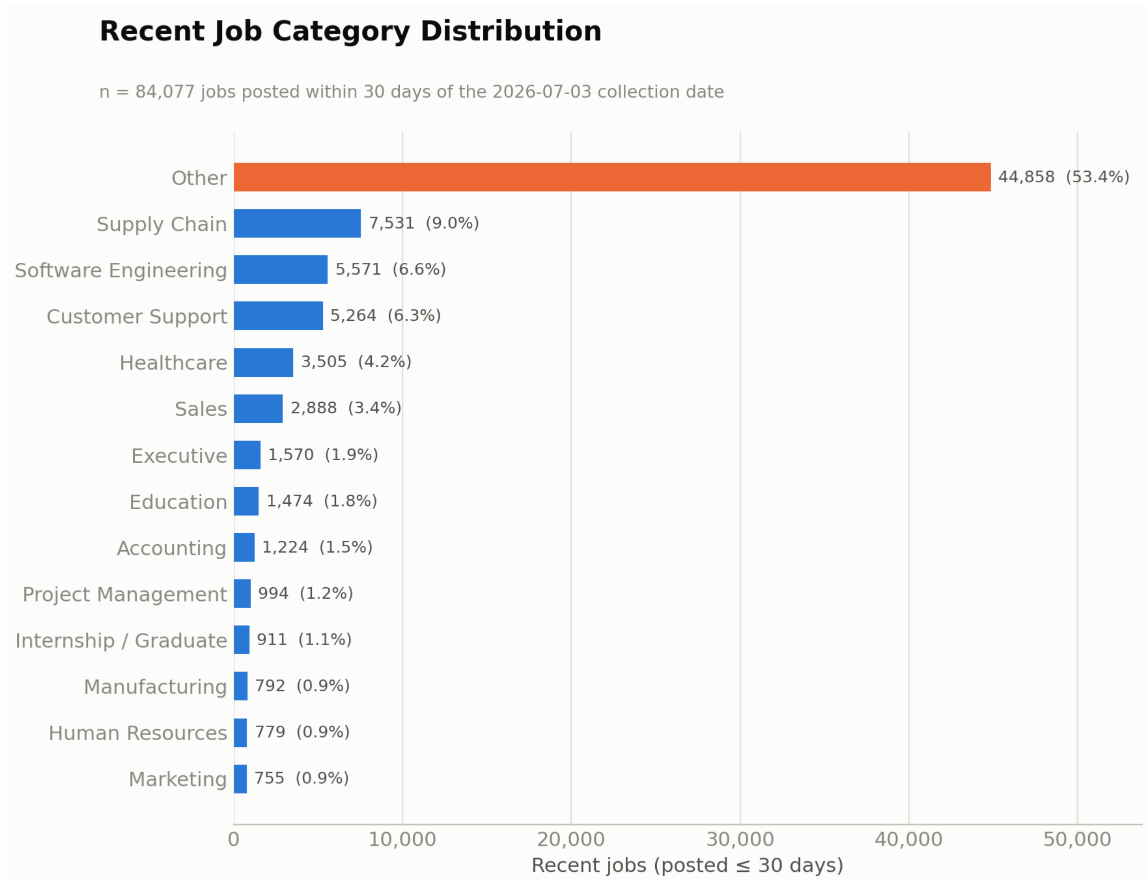

Figure 9. Recent job category distribution (posted within 30 days).

Excluding the Other catch-all, Supply Chain (9.0% — driven by delivery-driver and warehouse/despatch roles), Software Engineering (6.6%), and Customer Support (6.3%) are the three largest identifiable categories among recent postings, followed by Healthcare (4.2%) and Sales (3.4%). The Other bucket is dominated by retail, hospitality, and food-service titles ("Barista," "Tesco Colleague," "Kitchen Porter," "Assistant Manager") that fall outside the given taxonomy — a data-composition finding, not a classification gap.

2. Recent Jobs by Region

Region	Recent Jobs	% of recent
United States	39,124	46.5%
Europe	15,721	18.7%
Other	14,372	17.1%
United Kingdom	6,925	8.2%
LATAM	2,886	3.4%
Singapore / Southeast Asia	1,744	2.1%
India	1,656	2.0%
Canada	1,649	2.0%

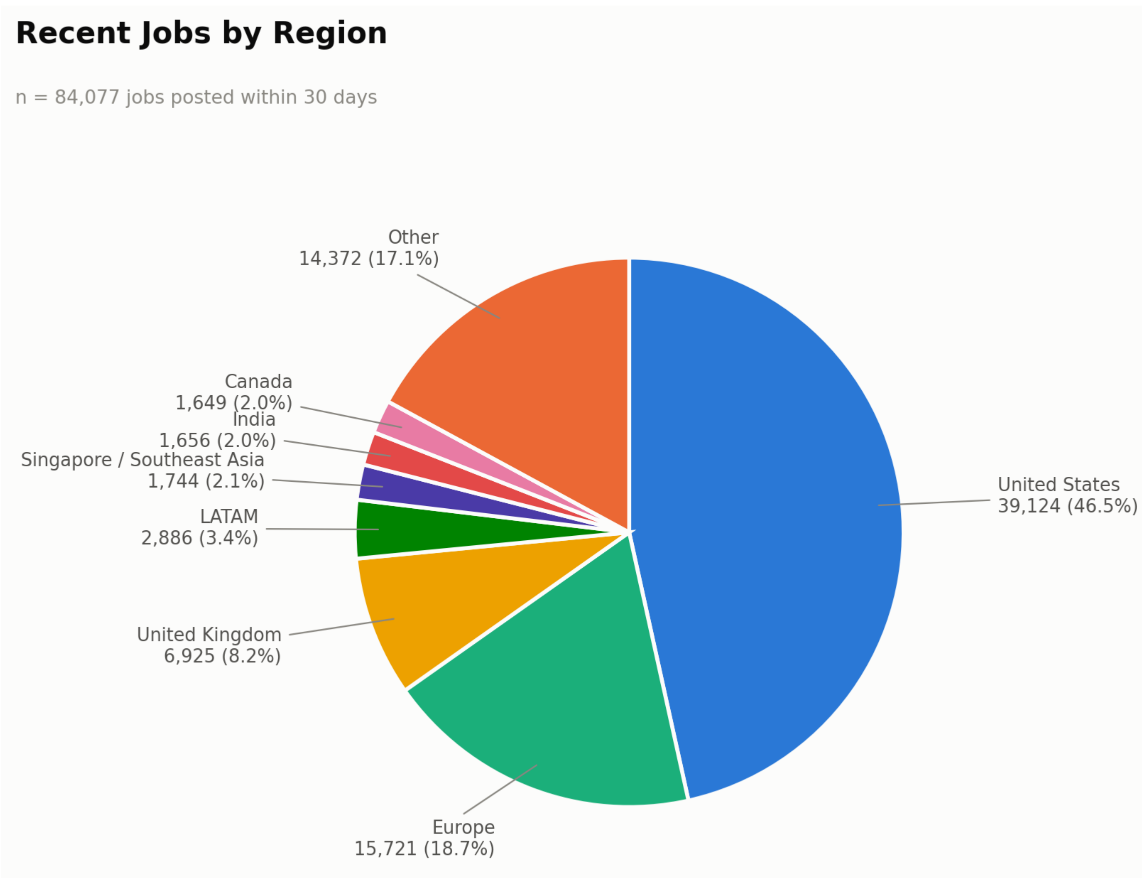

Figure 10. Recent jobs by region (posted within 30 days).

The United States dominates recent postings (46.5%), followed by Europe (18.7%, still Germany-heavy via Bundesagentur) and the UK (8.2%, almost entirely Reed, which is 96.2% fresh). Canada, India, and Singapore/Southeast Asia each sit at only ~2% of recent volume — these regions are thin in both total volume and freshness, reinforcing the coverage gaps identified in Section 6 and in research/api_analysis.md.

3. Recent Jobs by Provider

Provider	Recent Jobs	% of recent	Freshness rate (of provider's own total)
SmartRecruiters	36,479	43.4%	56.0%
Reed	8,658	10.3%	96.2%
Bundesagentur	7,800	9.3%	100.0%
USAJOBS	6,138	7.3%	61.4%
Other (SerpApi, OpenWeb Ninja, CareerJet, TheirStack, misc.)	5,091	6.1%	50.4%
Greenhouse	5,013	6.0%	33.6%
Jooble	2,973	3.5%	95.1%
Himalayas	2,964	3.5%	100.0%
Lever	2,653	3.2%	14.9%
Ashby	1,688	2.0%	32.6%
Adzuna	1,467	1.7%	73.3%
Workable	1,232	1.5%	29.6%
Arbeitnow	930	1.1%	100.0%
The Muse	615	0.7%	30.8%
Teamtailor	176	0.2%	38.6%
Jobicy	100	0.1%	100.0%
RemoteOK	100	0.1%	100.0%

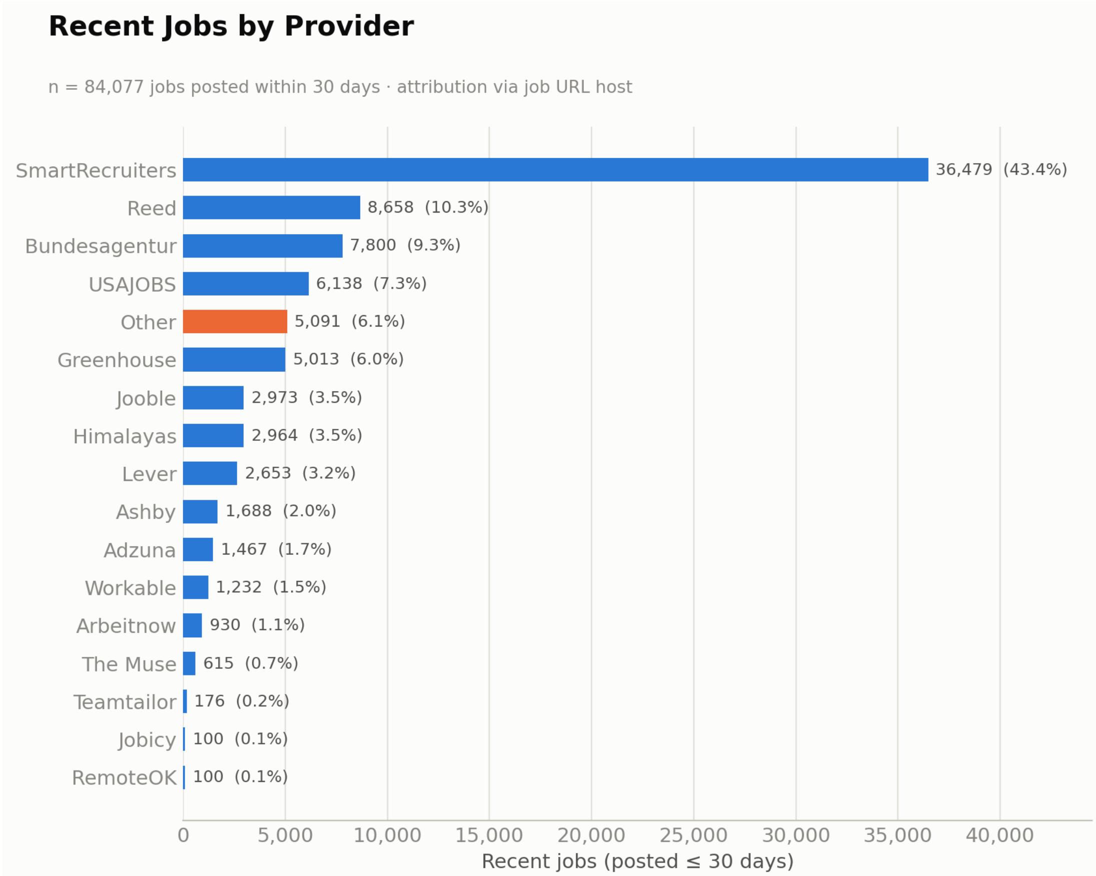

Figure 11. Recent jobs by provider (posted within 30 days).

By share of recent volume, SmartRecruiters (43.4%), Reed (10.3%), and Bundesagentur (9.3%) contribute the most fresh jobs in absolute terms. By freshness rate — the share of a provider's own total inventory that is recent, a more diagnostic metric — five providers are effectively 100% fresh: Bundesagentur, Himalayas, Arbeitnow, RemoteOK, and Jobicy, each a newest-first or short-window API by construction. Reed (96.2%) and Jooble (95.1%) are also excellent. Lever (14.9%), Workable (29.6%), The Muse (30.8%), Ashby (32.6%), and Greenhouse (33.6%) have the lowest freshness rates — consistent with ATS boards returning every open requisition regardless of age.

4. Recent Jobs by Company

Rank	Company	Recent Jobs	% of recent
1	Domino's	20,620	24.5%
2	AccorHotel	3,397	4.0%
3	AECOM	2,297	2.7%
4	Bosch Group	1,900	2.3%
5	Veterans Health Administration	1,675	2.0%
6	Anduril Industries	817	1.0%
7	Northwestern Memorial Healthcare	751	0.9%

Rank	Company	Recent Jobs	% of recent
8	Reed	689	0.8%
9	SpaceX	681	0.8%
10	Delivery Hero	599	0.7%
11	eFinancialCareers	514	0.6%
12	Primark	454	0.5%
13	ASSYSTEM	417	0.5%
14	Tesco	396	0.5%
15	AUREA GmbH	341	0.4%
16	Boyd Gaming	335	0.4%
17	Munson Healthcare	328	0.4%
18	NielsenIQ	324	0.4%
19	Hays Specialist Recruitment Limited	320	0.4%
20	Continental	309	0.4%

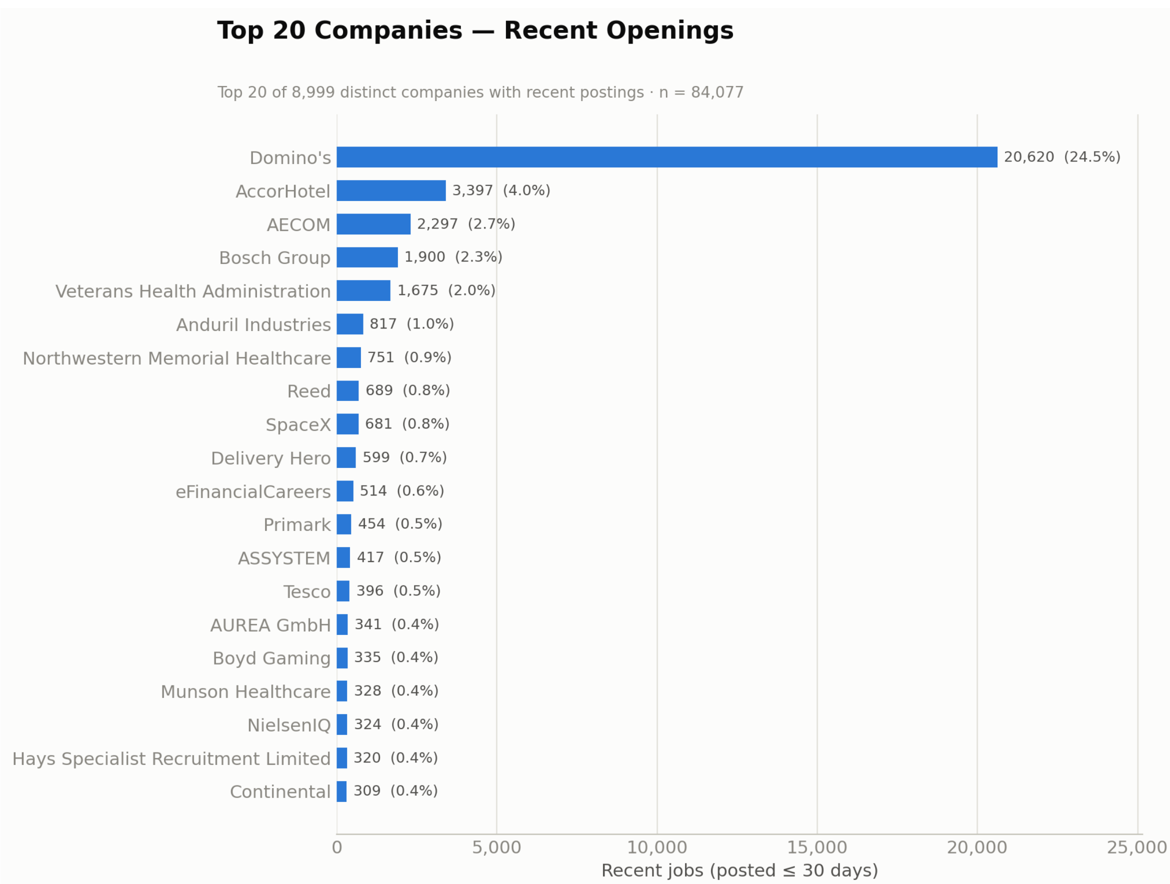

Figure 12. Top 20 companies by recent openings.

Recent hiring is more concentrated than the dataset as a whole: the top 10 companies account for 39.8% of recent postings (vs. 37.8% dataset-wide), and Domino's alone is 24.5% of every recent job — up from 15.7% dataset-wide — because Domino's postings refresh very frequently, while many other large employers' boards (particularly Lever- and Greenhouse-hosted companies) contain a larger share of older, still-open listings. There are 8,999 distinct companies among recent postings, and the same bimodal pattern holds as in the full dataset: a handful of very large, frequently-refreshed employers alongside a long tail of single-posting companies.

5. Fresh Job Recommendations

- APIs producing the freshest data: Bundesagentur, Himalayas, Arbeitnow, RemoteOK, and Jobicy are effectively 100% fresh — each is either newest-first within a small window or a small, frequently-turning-over snapshot by construction. Reed (96.2%) and Jooble (95.1%) are close behind.
- Providers that mostly return older jobs: Lever (14.9% fresh), Workable (29.6%), The Muse (30.8%), Ashby (32.6%), and Greenhouse (33.6%) — all five are full-board ATS fetches that return every currently open requisition regardless of posting age.
- Providers that should be refreshed more frequently: the same five low-freshness ATS providers benefit least from more frequent runs alone, since re-running them mainly re-fetches the same long-lived postings — the fix is incremental sync, not a tighter schedule.
- Providers that should be synchronized daily: SmartRecruiters, Reed, Bundesagentur, USAJOBS, and Jooble — all high-freshness-rate, high-volume providers whose own APIs are already turning over quickly enough to justify daily capture.
- Providers that would benefit from incremental synchronization: USAJOBS (DatePosted), Adzuna (max_days_old), Jooble (datecreatedfrom), Bundesagentur (veroeffentlichtseit), and TheirStack (posted_at_max_age_days, already partly used) all expose a genuine date-filter parameter today that is unused or only partly used.

6. Engineering Recommendations

- Date filtering: turn on the date filters already supported by USAJOBS, Adzuna, Bundesagentur, Jooble, and TheirStack to exclude stale postings at ingest time.
- Incremental synchronization: persist a per-job first_seen/last_seen watermark (using createdAt/publishedAt/releasedDate, already returned by every ATS provider) so runs can detect and retire postings that disappeared, rather than re-fetching a full board every time — the single biggest lever for Lever, Workable, The Muse, Ashby, and Greenhouse, none of which expose a request-side date filter.
- Region splitting: partitioning broad, single-query providers (Adzuna, Bundesagentur, USAJOBS, Reed) by country/region multiplies effective coverage and lets each partition be scheduled and filtered independently by recency.
- Keyword searches: replacing wildcard queries (Jooble, Careerjet, SerpApi, OpenWeb Ninja) with real, targeted keywords tends to surface different, often more recent, postings than a generic broad browse, while reducing cross-provider duplication.
- Company-based searches: for the six ATS providers, freshness cannot be improved via query parameters — there are none — so the lever is exclusively adding/pruning company slugs and diffing each company's board run-over-run to detect closed postings.

- Category searches: partitioning The Muse, USAJOBS, and Himalayas by category alongside date ordering would let each category-specific query surface its own newest postings rather than being crowded out by a single large unfiltered result set.

Estimated freshness improvement: applying date filters and incremental sync to the five providers identified above, and introducing run-over-run diffing for the five low-freshness ATS providers, would directly target the ~46,000 jobs in this dataset that are currently older than 30 days despite coming from providers capable of returning newer results. A realistic outcome is raising the overall <=30-day-fresh share from the current 54.0% into the high-60s to low-70s percent range, without adding a single new provider — the ceiling is bounded by how much genuinely new inventory each provider's underlying job market produces per period, not by anything in this pipeline.

18. Conclusion

The Job Aggregation Platform is a mature, well-engineered first version: a clean, extensible source architecture spanning 20 providers behind one shared interface, a fully normalized schema, correct URL-based deduplication, and disciplined per-source error isolation. It is production-ready as a periodic bulk job board, but not yet as a freshness- or scale-sensitive product — it remains a static snapshot with no scheduling, no expiry, and no persistent per-job state.

Regional strength is concentrated in the United States, Germany, and (for its size) the UK; Canada and Southeast Asia are thin, India has no dedicated source despite meaningful incidental volume, and the UK's real coverage rides on a single provider. Duplicate handling is excellent at the URL level, and the apparent duplication in title+company groupings is explained, not a defect. Freshness is the platform's weakest dimension and its highest-value fix, being primarily an architectural gap rather than a sourcing one.

Nearly all of the highest-impact opportunities identified in this report are already unlocked: multi-country Adzuna, MyCareersFuture, France Travail, ATS slug expansion, and India coverage via existing infrastructure are all free or low-cost, legally clean, and require no new commercial relationships.

On scalability: the current architecture cannot, as built, scale to millions of jobs. A single ~54 MB jobs.json at 156,000 jobs implies roughly 350 MB at 1 million and over 1 GB at 3 million — both the storage model and the frontend's whole-file load pattern would need to change. It can, however, comfortably grow into the low hundreds of thousands of jobs with the roadmap in this report, and the existing modular source design makes the eventual move to a datastore-backed, incrementally synced, queryable service a natural evolution rather than a rewrite.